Quantifying Socio-Linguistic Harm in Low-Resource Languages: A Conceptual Framework for Representational and Expressive Dimensions

R. Hoashalarajh and K. Sarveswaran
Department of Computer Science
University of Jaffna

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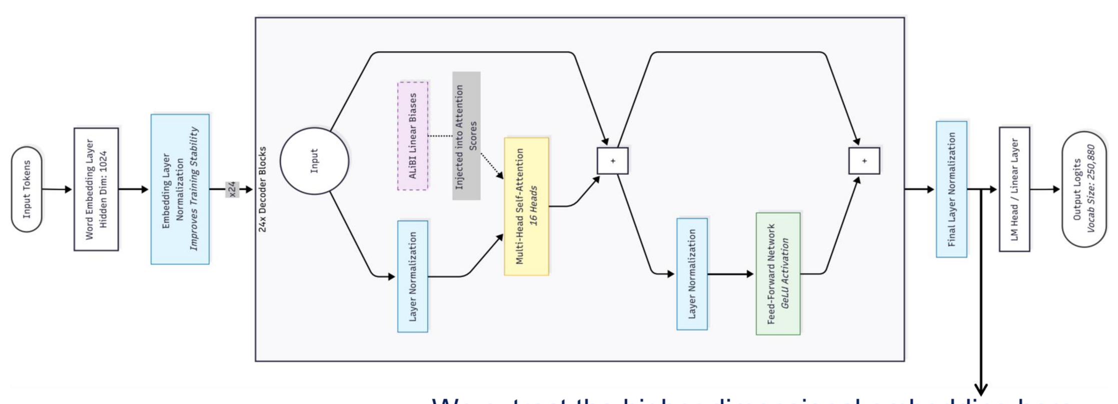
Introduction

Addressing harmful speech is vital because socio-linguistic discrimination inflicts severe psychological damage, reinforces systemic power imbalances, and silences marginalized groups.

The problems with current NLP systems are,

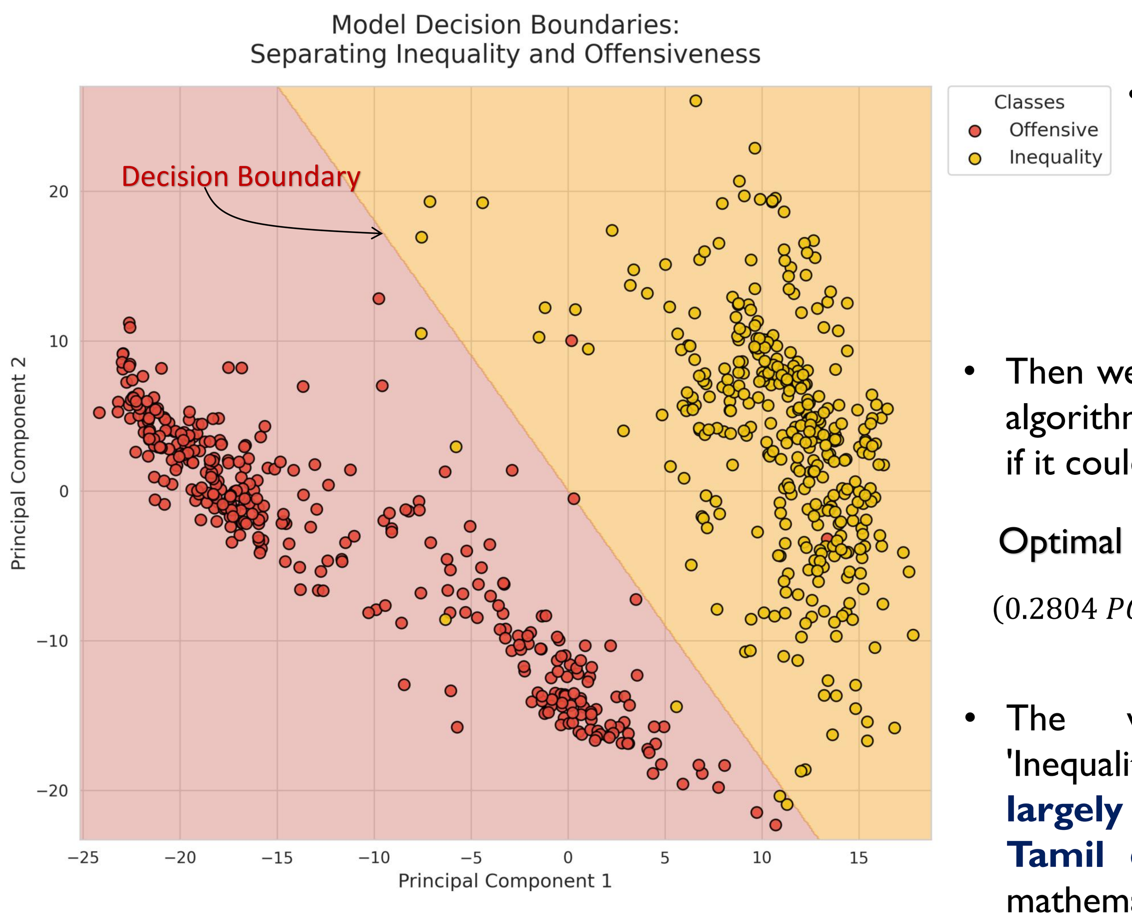
- **Reliance on Binaries:** Harmful speech detection is largely limited to binary labels (e.g., Hate vs. Non-hate) [1].
- **Missing the Nuance:** This fails to capture the interplay between what is said (Representational stance) and how it is conveyed (Expressive stance) [2].
- **Low-Resource Language Gap:** In morphologically rich languages like Tamil and Sinhala, this limitation is severe [3].

Motivation



We extract the higher dimensional embedding here

- We have fine-tuned the **BLOOM-560m** model for our downstream task of classifying Tamil texts under the categories of "Inequality", Offensive" and "Neutral".
- Trained on formal Tamil texts, **1 069** samples total size



- We **reduce the higher dimensional space to 2D space** by selecting **top two principal components**, where each text sits somewhere in the 2D space.
- Then we trained **support vector machine** algorithm on the **reduced 2D space** to see if it could find a **clear decision boundary.**

Optimal separating hyperplane:

$$(0.2804\, PC_1) + (0.1555\, PC_2) - 0.0089 = 0$$

- The visualization demonstrates that 'Inequality' and 'Offensiveness' inhabit **largely distinct semantic spaces within Tamil discourse**, allowing the model to mathematically separate them.

Research Questions

1. Can Socio-Linguistic Harm be modeled as a multidimensional latent construct composed of representational and expressive dimensions?
2. How can Representational and Expressive harm be computationally quantified in monolingual text?
3. How can representational harm arising from the erasure of socio-historical and inequality markers be identified and quantified across cross-lingual discourse?

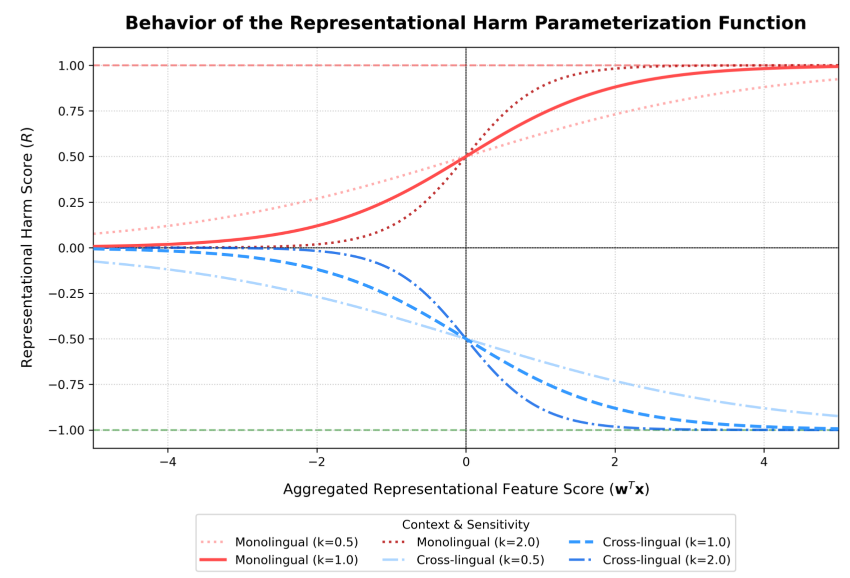
Methodology

Representational Harm: Language that promotes negative stereotypes, unequal portrayals, or erases socio-historical context of marginalized groups [3,4].

$$x(t) = [x_{hand}(t) \,||\, x_{emb}(t)]$$

$$x_{hand}(t) = \begin{bmatrix} stereotype\_score(t) \\ inequality\_score(t) \\ target\_group\_attack\_score(t) \\ negative\_sentiment\_score(t) \\ socio-historic\_erasure\_score(t) \\ caste\_gender\_intersection\_score(t) \end{bmatrix}$$

$$x_{emb}(t) = LaBSE(t); \quad z = w^T x(t); \quad f_{(z)} = \frac{1}{1+e^{-kz}}$$



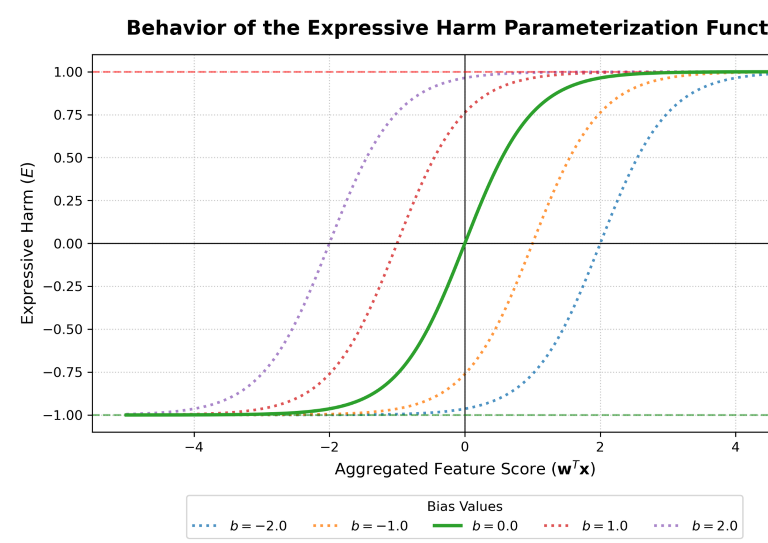
$$R_{(M,z)} = \begin{cases} f_{(z)}, & if\ M = monolingual \\ -f_{(z)}, & if\ M = cross\ lingual \end{cases}$$

Expressive Harm: The degree to which language conveys offense, derogation, or undermining, insulting intent toward individuals or groups [4,5].

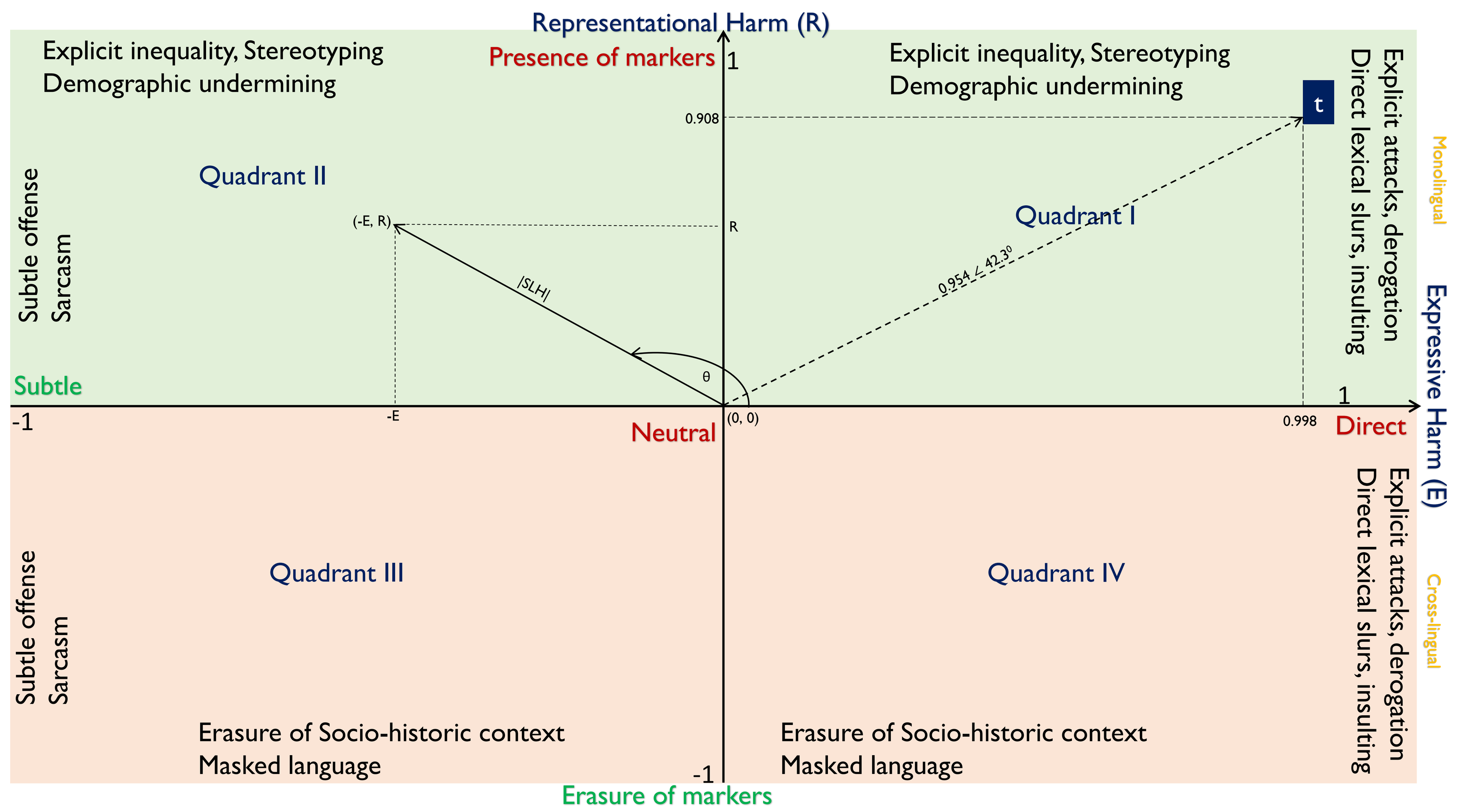
$$x(t) = [x_{hand}(t) \,||\, x_{emb}(t)]$$

$$x_{hand}(t) = \begin{bmatrix} abusive\_score(t) \\ amplifier\_score(t) \\ sarcastic\_score(t) \\ targeted\_attack\_score(t) \\ threat\_score(t) \\ repetition\_score(t) \\ emphasis\_score(t) \end{bmatrix}$$

$$x_{emb}(t) = LaBSE(t);$$



$$E(t) = tanh(w^T x(t) + b)$$



$$SLH = ||SLH|| \angle \theta$$

$$||SLH|| = \frac{\sqrt{E^2 + R^2}}{\sqrt{2}}$$

θ = atan2(R, E)

$0 \le \theta \le \pi$

$$SLH = ||SLH|| \angle \theta$$

$$||SLH|| = \frac{\sqrt{E^2 + R^2}}{\sqrt{2}}$$

θ = atan2(-R, E)

$0 \le \theta \le -\pi$

t = "அகதியாக வந்த இந்த வந்தேறிகள் , நமது வரலாற்றுச் சிறப்புமிக்க நிலங்களில் சட்டவிரோதமாகக் குடியேறி, அதன் புனிதத்தன்மையைக் கெடுப்பதுடன் நாட்டின் அமைதிக்கும் பெரும் சவாலாக உள்ளனர்; இவர்களை உடனடியாக இங்கிருந்து அடித்து விரட்ட வேண்டும்".

Translation: "These interlopers, who came as refugees, have illegally settled in our historically significant lands. In addition to desecrating their sanctity, they pose a major threat to the peace of the country; they should be immediately beaten and driven out from here".

Computing Expressive Harm:

$$x(t) = [0.9, 0.7, 0.0, 1.0, 0.9, 0.0, 0.6]^T$$
$$w = [1.0, 0.5, 0.0, 1.0, 1.0, 0.0, 0.5]^T$$
$$w^T x(t) = 3.45$$

Assuming $b = 0$

$$E(t) = tanh(w^T x(t) + b) = 0.998$$

Computing Representational Harm:

$$x(t) = [0.8, 0.9, 1.0, 0.9, 0.0, 0.0]^T$$
$$w = [0.5, 0.5, 1.0, 0.5, 0.0, 0.0]^T$$
$$z = w^T x(t) = 2.30$$

With k = 1.0

$$R_{(M,2.3)} = \frac{1}{1+e^{-2.3}} = 0.908$$

Socio-Linguistic Harm:

(E, R) = (0.998, 0.908)

||SLH|| = 0.954

$\theta = 42.3^0$

SLH = 0.954 ∠ 42.3⁰

Preliminary Experiment

To measure the divergence of information content in each text with respect to a reference text, we apply **Kullback-Leibler** divergence.

Here P is the reference distribution.
Q_1, Q_2 are two different texts and they are the target distribution.

$$D(P \parallel Q_1) = \sum_{x\in X} P(x)\big(\log(P(x)) - \log(Q_1(x))\big)$$

$$D(P \parallel Q_2) = \sum_{x\in X} P(x)\big(\log(P(x)) - \log(Q_2(x))\big)$$

$$\begin{aligned} D(P \parallel Q_1) - D(P \parallel Q_2) &= \\ &= \sum_{x\in X} P(x)\,[\cancel{\log(P(x))} - \log(Q_1(x)) - \cancel{\log(P(x))} + \log(Q_2(x))] \\ &= \sum_{x\in X} P(x)\,[\log(Q_2(x)) - \log(Q_1(x))] \\ &= \sum_{x\in X} P(x)\left[\log\frac{Q_2(x)}{Q_1(x)}\right] \end{aligned}$$

By definition of "Expected Value" the following expected log-likelihood ratio holds:

$$D(P \parallel Q_1) - D(P \parallel Q_2) = E_{x\sim P}\left[\log\frac{Q_2(x)}{Q_1(x)}\right]$$

Then we derived the expected log-likelihood ratio to compute the relative divergence of text Q_1 from Q_2 with respect to the assumed ground truth reference P.

Q1: මෙහිදී මන්නාරම ප්‍රදේශය නියෝජනය කරමින් පැමිණි ආගමික නායකයින් ඇතුළු ජනතාව කියා සිටියේ ඉල්මනයිට් ව්‍යාපෘතිය සහ සුළං බලාගාර ව්‍යාපෘතිය නිසා පරිසර හානි මෙන්ම ජන ජීවිතයට බලපෑම් එල්ල වනබවයි. කෙසේ වෙතත්, මෙම ව්‍යාපෘති අඩු පිරිවැයකින් විදුලිය නිෂ්පාදනය කිරීමසඳහා අවශ්‍ය ජාතික සම්පත් බවත්, ඒවා ප්‍රමාද කිරීමෙන් රටේ ආර්ථිකයට දැඩි හානියක් සිදුවන බවත් බලධාරීන් විසින් අවධාරණය කරන ලදී. (Hiru news, 13th of August 2025)

Q1, translation: Religious leaders and people representing the **Mannar** area stated that the Ilmenite project and the wind farm project are causing environmental damage and affecting people's lives. However, the authorities emphasized that these projects are critical national assets required to generate low-cost electricity and delaying them would severely harm the country's economy.

Q2: மன்னாரில் சிறுபான்மை தமிழ் மக்களின் உரிமைகளைப் புறக்கணித்து, அவர்களது பூர்வீக நிலங்களை அபகரித்து காற்றாலை மற்றும் கனியமணல் திட்டங்கள் திணிக்கப்படுவதே இங்குள்ள பிரதான பிரச்சினையாகும். அங்கே உள்ள சிறுபான்மை தமிழ் மக்களின் அனுமதியின்றி அவர்களின் வாழ்வாதாரங்களையும் அடிப்படை மனித உரிமைகளையும் அச்சுறுத்தும் இந்த அத்துமீறிய திட்டங்களை உடனடியாக நிறுத்தக் கோரியே இப்போராட்டம் முன்னெடுக்கப்படுகின்றது. (Midpoint.lk, 15th of August 2025)

Q2, translation: The main issue here is the imposition of wind farm and mineral sand projects, ignoring the rights of the minority **Tamil** people in **Mannar** and seizing their ancestral lands. This movement is being carried out to demand an immediate halt to these encroaching projects that threaten the livelihoods and basic human rights of the minority Tamil people there without their consent.

P, reference: A mass protest in **Mannar**, organized by local **Tamil** civil groups, demanded the immediate halt of wind power and mineral sand mining projects. Demonstrators argued these operations cause environmental degradation, facilitate land grabs without public consent, and threaten the fundamental rights of the minority local Tamil community. (Tamil Guardian, 12th of June 2025)

Entities Identified in Reference (P):
- 'Mannar' [GPE]: Geopolitical Entity
- 'Tamil' [NORP]: Nationalities, Religious groups, and Political groups.

Entities Identified in Sinhala (Q1):
- 'Mannar' : MATCHED
- 'Tamil' : UNMATCHED

Entities Identified in Tamil: (Q2)
- 'Mannar' : MATCHED
- 'Tamil' : MATCHED

Metric	Comparison	Outcome (nats)
$D(P \parallel Q_1)$	Reference (P) vs. Sinhala-Translated (Q_1)	5.3812
$D(P \parallel Q_2)$	Reference (P) vs. Tamil-Translated (Q_2)	0.3298
Relative Divergence	$\mathbf{E_{x\sim P}}\left[\log\frac{\mathbf{Q_2(x)}}{\mathbf{Q_1(x)}}\right]$ (> 0)	**5.0514**

- Relative Divergence is measured in **natural** units (nats) in base e.
- **In numbers:** Text Q1 diverges from the original information content **156.24 times more than** Q2 with respect to the reference text P.

Conclusion and Discussion point

- The proposed Socio-Linguistic Harm (SLH) framework moves beyond monolithic "hate vs. non-hate" labels by modeling linguistic harm as a continuous, two-dimensional geometric latent construct.
- By separating Representational Harm (R) and Expressive Harm (E), the polar metric (SLH = ||SLH|| ∠ θ) effectively quantifies both overt explicit attacks and the subtle, systemic erasure of socio-historic markers.
- D: What are your thoughts on establishing ground truth when multiple news sources present biased or conflicting narratives?

References
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[4] Gallegos, Isabel O., et al. "Bias and fairness in large language models: A survey." Computational linguistics 50.3 (2024): 1097-1179.
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